LandEmu: a learned emulator of the trait-based, individual-tree vegetation model LPJmL-FIT

Daily carbon and water exchange and forest structure at a fraction of the cost

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Manuscript draft, 22 October 2026

Abstract

Individual-based, trait-based vegetation models resolve how forests of differently adapted trees grow, compete and die, but they are too expensive to run inside Earth system models. We present LandEmu, a learned emulator of the dynamic global vegetation model LPJmL-FIT that reproduces both its daily carbon and water exchange and its yearly forest structure and trait composition, driven by daily weather alone. Profiling the original shows that its cost lies in the daily per-tree physics and in the ensemble of replicate patches it averages over, so the emulator replaces the daily step as a whole and predicts the expectation over patches. A five-member ensemble of small neural networks computes the daily fluxes, with snow, top-metre and deep soil water carried as explicit stores that close the water balance exactly. Once a year, gradient-boosted tree models predict tree density, biomass per tree, four trait medians, leaf area, vegetation carbon and plant-type cover as a departure from a per-cell anchor, driven by the trailing 30-year climate, and a second boosted model adds the year-to-year departure of tree density and biomass per tree driven by the year's weather anomaly. Tested on a climate model and an independent run of LPJmL-FIT withheld from training, the yearly model's error in dense forests is 1.22 (tree density) and 1.18 (biomass per tree) times that of a further run of the stochastic original, and 0.59–0.86 times for the trait medians. The error grows by a factor of 1.05 between the 2050s and the 2090s. In the original's drought die-offs, which are threshold kills of individual trees, the emulator follows 18 % of the tree loss. Tested on cells withheld from training, the daily networks' 30-year mean fluxes are 2.4 (gross primary production) to 5.2 (runoff) times further from the original than a further run is — about 3–6 % per cell for production and evapotranspiration — with area totals within 3 %. For a nitrogen-limited configuration of LPJmL-FIT with rising CO_2, the emulator reproduces 81–97 % of the change in vegetation carbon under two unseen climate models (a second run of the original: 61–89 %), but only 72 % of the largest warming-only loss, and it damps the cell-to-cell spread of the response. The emulator costs 7.0×10^{-3} core-seconds per cell-year, about 38 and 1 000 times less than LPJmL-FIT at 25 and 500 patches.

1 Introduction

Terrestrial ecosystems take up about a third of anthropogenic CO_2 emissions (Friedlingstein et al., 2023), and the vegetation that does so is not a fixed surface but a population of plants that grow, compete, die and are replaced. Dynamic global vegetation models (DGVMs) represent this population in Earth system models (ESMs), usually through a few plant functional types with fixed properties. A growing body of work argues that this is not enough: the size structure and the functional diversity of forests shape how they respond to drought and warming, and models that resolve individuals, cohorts and traits behave differently from those that do not (Fisher et al., 2018; Fisher and Koven, 2020). LPJmL-FIT, a version of the LPJmL model with individual trees and variable traits (Sakschewski et al., 2015, 2016; Thonicke et al., 2020), is one of these models: each tree carries its own specific leaf area, wood density, rooting depth and drought threshold, drawn when it establishes, and the forest's trait composition shifts as trees with poorly suited traits die. Amazon forests with more trait diversity recover better from drought in this model than forests without it (Sakschewski et al., 2016).

The price of this realism is cost. LPJmL-FIT simulates every tree every day, and because establishment and mortality are random it averages over many replicate patches per grid cell: 25 in the runs used here, around 500 in published applications. Its run time grows in proportion to the number of patches, which puts it orders of magnitude beyond what an ESM can afford for its land surface. Emulation — replacing the expensive model with a fast statistical or machine-learned approximation — is the obvious route.

Emulators of land models. Statistical emulation of land models began as a tool for calibration and sensitivity analysis. Gaussian-process and polynomial-chaos surrogates map a model's parameters to a few outputs or to a cost function, so that the expensive model need not be run for every parameter set (Sargsyan et al., 2011; Fer et al., 2018; Baker et al., 2022; Dagon et al., 2020); history matching and Bayesian calibration of JULES, ORCHIDEE and TRIFFID work this way (McNeall et al., 2024; Raoult et al., 2024; Hawkins et al., 2019). Such surrogates answer how a summary of a run depends on parameters, not how the land state evolves under a changing atmosphere, and they cannot stand in for a land component. Since about 2019 deep learning has widened what an emulator can be: recurrent, convolutional and generative networks map climate forcing to multivariate trajectories, and differentiable programming lets learned components be trained inside process models (Reichstein et al., 2019; Shen et al., 2023).

A few emulators now reproduce a land model broadly enough to replace part of it. For ECMWF's land scheme ecLand, LSTM, gradient-boosted and feed-forward emulators of seven prognostic soil and snow states run 44 to 4800 times faster than the model (Wesselkamp et al., 2025), and aiLand adds the turbulent heat fluxes, predicts state increments and runs stably for four years (Raoult et al., 2026). For the demographic model ED, a multi-scale LSTM emulates canopy height, biomass, leaf area, production and soil respiration over 40-year trajectories (Wang et al., 2023), and a conditional diffusion model emulates the same outputs probabilistically (Li et al., 2026). A statistical emulator of LPJ-GUESS carbon, forestry and crop outputs serves integrated assessment (Ekholm et al., 2024), forest-carbon emulators of LPJ-GUESS predict grid-cell stocks (Natel et al., 2025), PHASE predicts the equilibrium of slow carbon pools of the E3SM land model to shorten spin-up by a factor of 60 (Gao et al., 2025), and a physically guided emulator of the MATSIRO land surface model keeps its water budget (Olson et al., 2024). Statistical emulators of LPJmL carbon fluxes and crop yields have been built for coupling with climate and economic models (Oyebamiji, 2015; Oyebamiji et al., 2015), and a canopy-temperature emulator runs inside LPJmL's crop module (Hemmen et al., 2026). Each of these covers part of a land model: soil water and heat with turbulent fluxes, or carbon and stand structure, or equilibrium pools. None emulates the size and trait distributions of a forest or the mortality of individual trees; the closest is a state-transition network for the landscape model iLand (Rammer and Seidl, 2019).

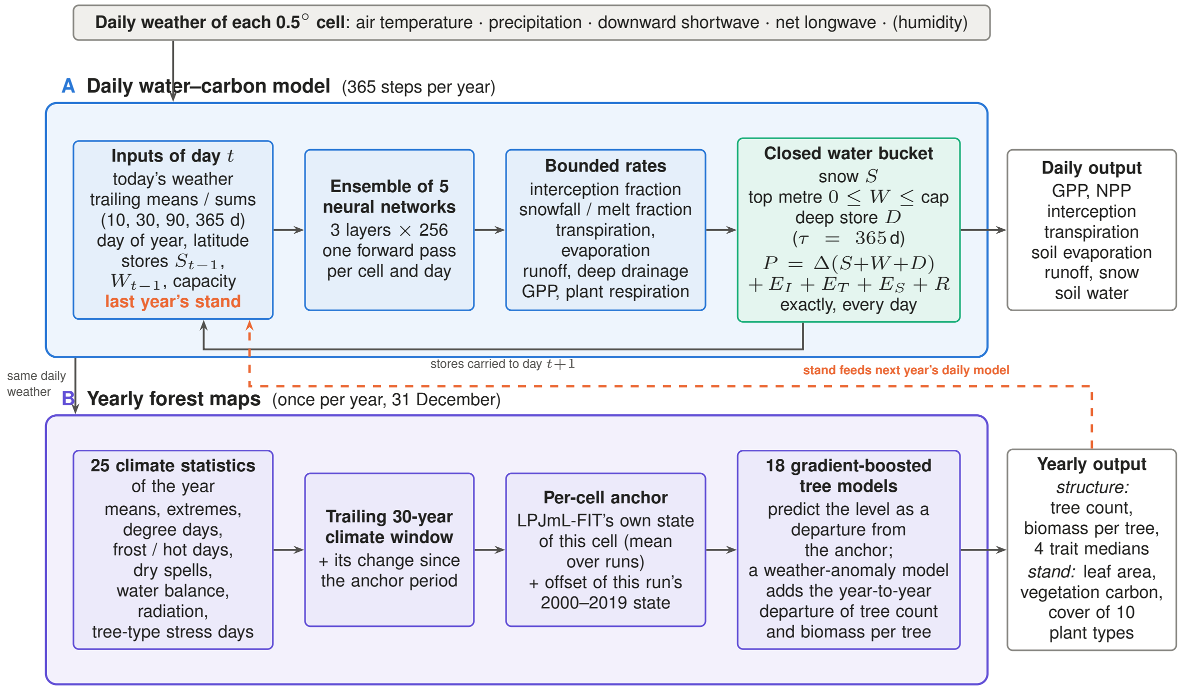
The second route keeps the process model and replaces individual processes with learned components. Fire is the most common target: machine-learned fire models have replaced the process schemes of the E3SM land model and of JSBACH (Zhu et al., 2022; Liu et al., 2024, 2025; Son et al., 2024). Learned stomatal conductance and photosynthetic capacity run inside JSBACH within ICON-ESM (ElGhawi et al., 2025), learned snow cover improves a climate model over High Mountain Asia (Lalande et al., 2023), and an LSTM replaces the groundwater solver of a coupled land–subsurface model (Yang et al., 2026). Differentiable land models learn parameters or process functions end to end from observations (Aboelyazeed et al., 2023; Xu et al., 2026; Lahlou et al., 2026; Jiang et al., 2026; Fang et al., 2026), and hybrid water–carbon models couple neural networks with conceptual balance equations (Baghirov et al., 2025, 2026). These hybrids inherit the process model's structure and conservation, and they are so far the only learned components that run online inside the land component of an ESM — but they leave most of the computational cost in place.

What goes wrong. The same literature shows where emulators fail. Tree ensembles trained to predict one step ahead drift when run freely at long lead times, while networks trained on their own multi-step rollouts do not (Wesselkamp et al., 2025; Raoult et al., 2026): an emulator that carries its own state must be robust to its own errors, a problem known from imitation learning (Ross et al., 2011). Agreement in the present climate says little about agreement under a changed climate: an emulator of the ecosystem model ecosys matched present-day net primary production almost exactly yet underestimated it by $204\,\mathrm{gC\,m^{-2}\,yr^{-1}}$ by 2100 (Shirley et al., 2023), and an LPJ-GUESS emulator trained on stylised climate perturbations failed to transfer to realistic scenarios (Natel et al., 2025). Offline skill can collapse once feedbacks act: the spatial correlation of a machine-learned fire model in the E3SM land model fell from 0.96–0.98 offline to 0.59–0.70 online (Liu et al., 2024, 2025), and an online deep-learning fire model in JSBACH stayed below its offline skill (Son et al., 2024). An LSTM groundwater solver ran stably online but accumulated error late in the run (Yang et al., 2026). Water and carbon budgets are closed reliably only when the architecture enforces them rather than a loss term encouraging them (Baghirov et al., 2025, 2026). Tests under climates outside the training range, long free-running tests and explicit conservation checks remain the exception rather than the rule.

From this evidence follow the properties an emulator must have to stand in for an ESM's land component: prognostic state that carries the memory of past forcing; outputs that cover what the atmosphere exchanges with the land; conservation by construction; stability when run freely and under feedback; credibility under climates outside the present one; and quantified uncertainty with a software interface a host model can call. No published emulator of a dynamic vegetation model combines them, and none of the full emulators has been coupled to an ESM.

This paper. We describe LandEmu, an emulator of LPJmL-FIT designed against these properties. Its contributions are:

1. a complete emulator of an individual-tree, trait-based DGVM (Fig. 1) — daily gross and net primary production, interception, transpiration, soil evaporation, runoff, snow and soil water, together with yearly tree density, biomass per tree, four trait medians, leaf area, vegetation carbon and plant-type cover — that runs from daily weather alone and conserves water exactly;
2. a design derived from where the original spends its run time and from what the literature shows to fail, with every architectural and training choice backed by a measured comparison (Sects. 3, 4.5);
3. an evaluation that holds out a whole climate model and an independent run of the original (yearly maps) or whole blocks of cells (daily networks), and that measures every error against the original's own run-to-run variability (Sect. 5);
4. an analysis of the original's drought die-offs and of how far a stand-level emulator can follow them;
5. a test of whether the emulator learns the warming response when the training climates cover the test climate, a full-grid application, and an end-to-end cost measurement on one core against the original at its training and publication settings.


Daily weather of each 0.5° cell: air temperature · precipitation · downward shortwave · net longwave · (humidity)
A Daily water–carbon model (365 steps per year)
Inputs of day t
today's weather
trailing means / sums
(10, 30, 90, 365 d)
day of year, latitude
stores S_{t-1},
W_{t-1}, capacity
last year's stand
Ensemble of 5 neural networks
3 layers × 256
one forward pass
per cell and day
Bounded rates
interception fraction
snowfall / melt fraction
transpiration,
evaporation
runoff, deep drainage
GPP, plant respiration
Closed water bucket
snow S
top metre $0 \le W \le$ cap
deep store D
($\tau = 365$ d)
$P = \Delta(S+W+D)$
$+E_I + E_T + E_S + R$
exactly, every day
Daily output
GPP, NPP
interception
transpiration
soil evaporation
runoff, snow
soil water
stores carried to day $t+1$
stand feeds next year's daily model
same daily weather
B Yearly forest maps (once per year, 31 December)
25 climate statistics
of the year
means, extremes,
degree days,
frost / hot days,
dry spells,
water balance,
radiation,
tree-type stress days
Trailing 30-year climate window
+ its change since
the anchor period
Per-cell anchor
LPJmL-FIT's own state
of this cell (mean
over runs)
+ offset of this run's
2000–2019 state
18 gradient-boosted tree models
predict the level as a
departure from
the anchor;
a weather-anomaly model
adds the year-to-year
departure of tree count
and biomass per tree
Yearly output
structure:
tree count,
biomass per tree,
4 trait medians
stand: leaf area,
vegetation carbon,
cover of 10
plant types

State carried from year to year: the three water stores, the trailing weather windows, the 30-year climate window, the stand vector and last year's departures of tree count and biomass per tree.
Run-time dependencies: numpy and LightGBM. **Cost:** 7.0×10^{-3} core-s per cell-year, 97 % in block A.

Figure 1. How LandEmu advances one year. (A) The daily water–carbon model: the inputs of day t pass through an ensemble of five neural networks whose bounded outputs drive a closed three-store water bucket; the stores are carried to the next day. (B) The yearly forest maps: 25 climate statistics of the year enter a trailing 30-year window, and 18 gradient-boosted tree models predict the forest structure and stand as a departure from a per-cell anchor; a further boosted model adds the year-to-year departure of tree density and biomass per tree from the year's weather anomaly. The stand (leaf area, vegetation carbon, cover of ten plant types) is the daily model's input in the following year (dashed).

2 The model to be emulated: LPJmL-FIT

LPJmL-FIT couples the water, carbon and vegetation dynamics of LPJmL (Schaphoff et al., 2018) with an individual-tree forest module (Sakschewski et al., 2015; Thonicke et al., 2020). On each 0.5° grid cell the forest grows on a set of replicate patches (225 m^2 each). Trees of seven woody plant types establish with traits drawn within type-specific ranges, or inherited with small variations from large trees of the same patch; they compete for light in a layered canopy and for water in a 23-layer soil, allocate carbon each year, and die from carbon starvation, age, heat and drought stress, or are removed when their leaf carbon falls below that of a sapling. Three grass types fill the remaining space. The daily step computes photosynthesis and stomatal conductance per tree, interception, transpiration, soil evaporation, snow, infiltration, percolation and runoff; the yearly step allocates carbon, applies mortality and fire, and establishes new trees. We use the configuration with natural vegetation only. For most of this paper nitrogen limitation is switched off and atmospheric CO$_2$ is held constant in the future runs (at its 2019 level; 2014 in the runs of Sect. 6.7); this is how LPJmL-FIT is usually run, because without nitrogen limitation its CO$_2$ fertilisation is unbounded. The emulator of this configuration therefore takes no CO$_2$ input, which is faithful to the original rather than a gap. Section 6.6 uses a second configuration, with nitrogen limitation (von Bloh et al., 2018) and rising CO$_2$, for which the emulator receives CO$_2$ as an input.

Where the run time goes. We profiled the unmodified production binary at five contrasting cells (temperate, tropical, semi-arid, Mediterranean, boreal; 25 patches, one core) and attributed the exclusive run time to processes, assigning the statically linked mathematical library calls

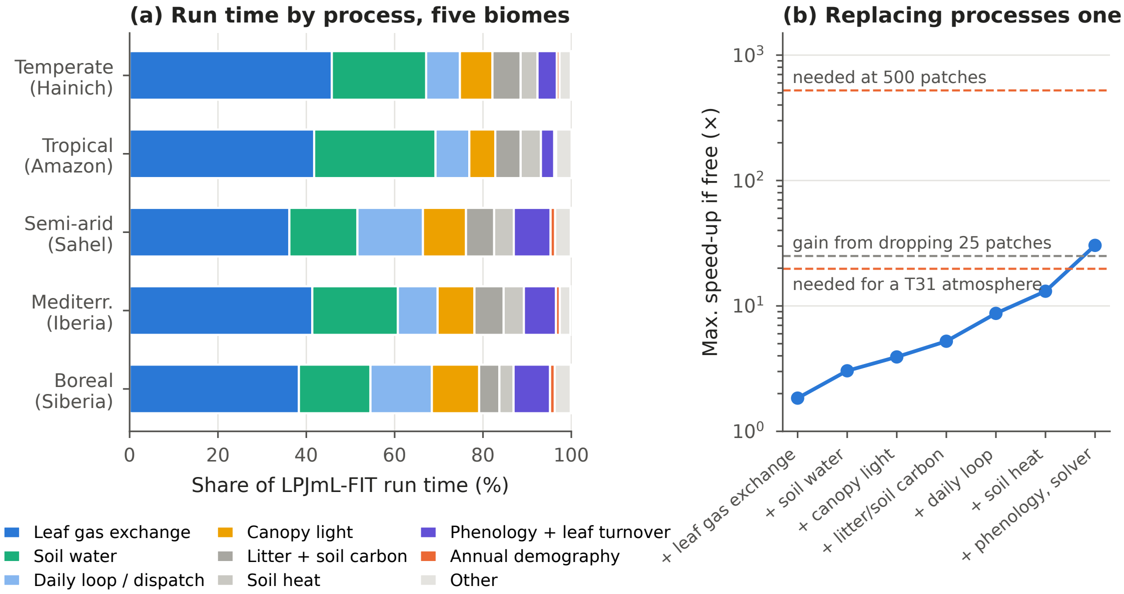


Figure 2. Where LPJmL-FIT spends its time. (a) Exclusive run time by process at five biome cells (25 patches, one core; mathematical library calls attributed to their callers). (b) The largest speed-up obtainable at the temperate cell if processes were made free of cost one after another, against the speed-up needed for an intermediate-resolution (T31) atmosphere, the gain from dropping the 25-patch ensemble, and the speed-up needed at 500 patches.

(21–27 % of all time) to the routines that call them (Fig. 2a). Leaf gas exchange takes 36–46 % of the run time and soil water another 15–27 %. The yearly demography — allocation, mortality, establishment and turnover — takes only 0.4–1.1 %, and the daily loop as a whole 97–98 %. The marginal cost is 0.20–0.33 core-seconds per cell-year at 25 patches, and it grows linearly with the number of patches. Even if the eight most expensive processes were free of cost, the model would speed up at most 30-fold at the temperate cell (Fig. 2b), whereas an intermediate-resolution atmosphere leaves an allowance of about 0.0135–0.030 core-seconds per land cell-year (10 % of a measured coupled atmosphere's cost at T31 and T63 resolution). Removing the 25-patch ensemble alone is worth a factor of 25, and of about 500 at the publication setting.

The original is stochastic. Two independent runs of LPJmL-FIT — each with its own 1000-year spin-up and random seed, under identical climate — disagree on a typical cell by about 6 % in the number of trees taller than 5 m, 12 % in biomass per tree, 2–10 % in the trait medians and 1.0–1.6 % in 30-year mean fluxes. In sparse stands the disagreement is far larger. Any emulator can at best predict the expectation over such runs; this sets both the target and the yardstick of the evaluation (Sect. 5).

3 Design

Four observations fix the architecture.

(i) The cost is the daily step and the patch ensemble (Sect. 2). No process of the original takes enough run time that accelerating it alone would give an order of magnitude, so a substantial speed-up must replace the daily step as a whole, and it must predict the expectation over patches instead of simulating replicates. Re-implementing physics can be faster than legacy code for a column model without a per-individual loop (Lahlou et al., 2026); for an individual-tree model the per-tree daily step is the cost.

(ii) One-step learning of the forest state drifts when run freely. A distributional random forest trained to predict the number of trees per patch one year ahead reached an out-of-sample

Table 1. What LandEmu reads, carries and returns.

Daily input	air temperature, precipitation, downward shortwave, net longwave (all days); specific humidity (yearly statistics only); CO_2 in the nitrogen-on variant
Per-cell set-up	latitude, longitude, soil code, top-metre plant-available water capacity; anchors of every yearly quantity; the climate of the anchor runs; the run's own 2000–2019 state and its offset from the anchor; initial stores, stand and weather windows; yearly climate statistics of the preceding years
State carried	snow, top-metre water, deep water; trailing weather sums and means (10, 30, 90, 365 days); 30-year window of yearly climate statistics; last year's stand; last year's departures of tree density and biomass per tree
Daily output	gross primary production (GPP), net primary production (NPP), interception, transpiration, soil evaporation, runoff, snow water equivalent, top-metre soil water, deep store
Yearly output	trees taller than 5 m per patch, above-ground biomass per tree, medians of specific leaf area, wood density, maximum rooting depth and drought threshold; stand leaf area index, vegetation carbon, fractional cover of ten plant types

coefficient of determination of 0.982 — but carrying the previous year's count forward already reached 0.962. Fed its own output, its area-weighted warming response turned from +0.71 to −0.23 of the original's within decades, because it followed 96 % of large increases but only 87 % of large decreases: a small, state-dependent asymmetry that accumulates. This is the failure reported for one-step tree ensembles of ecLand (Wesselkamp et al., 2025).

(iii) One climate per place cannot teach a response. Across the global grid, 76 % of a cell's warming is predictable from its present climate, so with one historical and one future run the climate response is confounded with geography — the identification problem behind the out-of-distribution failures in the literature (Shirley et al., 2023; Natel et al., 2025). The training data must contrast many climates at the same place.

(iv) Water and carbon are exchanged through stores. Conservation is reliable when the architecture enforces it (Baghirov et al., 2025, 2026), and the stores must be prognostic for an atmosphere model to exchange water with them.

From these follow the design principles of LandEmu. **P1**: learn the whole daily step with one small network per cell and day, keeping the water stores explicit and closed. **P2**: predict the expectation over the patch ensemble, so that the number of patches drops out of the cost. **P3**: predict each year's forest state as a departure from a per-cell anchor, driven by climate, rather than from last year's state, so that errors cannot accumulate; carry from year to year only a bounded, weather-driven departure from that level (Sect. 4.4). **P4**: train on climate contrast — many climate models and scenarios and several independent runs — and test on a climate model and a run never seen. **P5**: measure every error against the original's own run-to-run variability.

4 Emulator description

LandEmu advances one year at a time in two blocks (Fig. 1; Table 1). Block A, the daily water–carbon model, runs 365 daily steps from the day's weather and last year's stand. Block B, the yearly forest maps, runs once at the end of the year from climate statistics of the same weather and predicts the forest structure and the stand, which becomes the daily model's input for the next year. Everything is driven by daily weather on the 0.5° grid: air temperature, precipitation, downward shortwave radiation and net longwave radiation; specific humidity enters only the yearly climate statistics. All years have 365 days.

4.1 Daily water–carbon model

Inputs. For cell c and day t the network sees today's weather; trailing means of temperature (10 and 30 days) and shortwave radiation (30 days) and trailing sums of precipitation (30, 90 and 365 days), which stand in for the memory of the deep soil without carrying it as a state; the day of year (as sine and cosine), the sine of latitude and the day length; yesterday's snow S_{t-1} and top-metre water W_{t-1}, W_{t-1} divided by the capacity, and the capacity itself; and the stand at the end of the previous year (leaf area index, vegetation carbon and the cover of the ten plant types). No input is taken from the fluxes of the run being emulated. The top-metre water capacity is recomputed by LPJmL-FIT from its evolving soil carbon; in the evaluation runs it is taken each year from the run being emulated, in the released emulator it is fixed at each cell's 2019 value. Over the century it changes by a median 0.1–0.2 % (at most 4–6 % in single cells), so the difference is immaterial. The soil enters only through this capacity: soil texture, which sets how fast water enters and moves through the soil, is not an input (Sect. 4.5).

Network. Each of five ensemble members is a multilayer perceptron with three hidden layers of 256 units and SiLU activations; all outputs share one trunk. The outputs are mapped to bounded rates: an interception fraction $i \in [0, 1]$, a snowfall fraction $f \in [0, 1]$ and a melt fraction $m \in [0, 1]$, non-negative transpiration E_T, soil evaporation E_S and runoff R, a signed exchange Q between the top metre and the deep soil, non-negative GPP and plant respiration R_a, so that NPP = GPP − R_a. The ensemble mean of the five members is used.

Closed water bucket. With precipitation P_t, the stores are updated as

$$S_t = S_{t-1}(1 - m) + fP_t, \qquad E_{I,t} = iP_t, \tag{1}$$
$$W_t = W_{t-1} + P_t - (S_t - S_{t-1}) - E_{I,t} - E_{T,t} - E_{S,t} - R_t - Q_t, \tag{2}$$
$$D_t = (D_{t-1} + Q_t)(1 - 1/\tau), \tag{3}$$

where D is a deep store that drains the amount $(D_{t-1} + Q_t)/\tau$ into runoff each day, with a time constant $\tau = 365$ days. Water above the top-metre capacity is added to runoff and a deficit below zero is taken from the deep store. Hence, every day and to rounding precision,

$$P_t = \Delta(S + W + D)_t + E_{I,t} + E_{T,t} + E_{S,t} + R_t^{\mathrm{tot}}, \tag{4}$$

with R^{tot} the total runoff. The bounds restrict nothing the original does: on eight million cell-days interception and snow accumulation never exceeded precipitation and no flux turned negative. Because the deep store can be drawn below zero, its drainage term, and with it total runoff, turns slightly negative on a few days (minimum −2.5 to −10 mm per day in the cells tested).

Training. The networks are trained one step ahead (yesterday's stores from the original) on 40 million sampled cell-days of 730 training cells, with 80 further cells for early stopping, minimising the sum over the eight targets of the squared errors standardised by each target's spread (Adam with a one-cycle learning-rate schedule peaking at 3×10^{-3}, batches of 16 384, ten epochs; about 15 minutes per member on one GPU). They are then run freely: the stores are carried by the model over 81 years. Specific humidity is not an input, because the original's daily exchange does not read it.

4.2 Yearly climate statistics

At the end of each year, LandEmu computes 25 statistics of the year's daily weather: mean, coldest-month and warmest-month temperature; degree days above 5 °C; frost days and days above 30 °C; annual and summer precipitation; potential evaporation; the annual, summer and driest three-month climatic water balance; the longest dry spell; mean relative humidity; summer vapour pressure deficit and its 10-day extremes; mean shortwave and net longwave radiation; and, for each of the seven tree types, the number of days outside its temperature-stress range. These enter a trailing 30-year window, whose means are the climate the forest maps see.

4.3 Yearly forest maps

For each yearly quantity q (six structure quantities and twelve stand quantities) a separate gradient-boosted tree model (Ke et al., 2017) predicts

$$M^q_{c,y} = A^q_{c,y} + g_q\Big(\bar{x}_{c,y},\ \bar{x}_{c,y} - \bar{x}^A_{c,y},\ A_{c,y},\ s^0_c,\ \Delta s^0_c,\ \phi_c, \lambda_c, \sigma_c\Big)\,, \tag{5}$$

where $A^q_{c,y}$ is the *anchor*: the original's mean of q at cell c in year y over its training runs and climates; $\bar{x}_{c,y}$ is the trailing 30-year climate and $\bar{x}^A_{c,y}$ the corresponding climate of the anchor runs, so that their difference measures how far this climate departs from what the anchor represents; $A_{c,y}$ collects all anchors of the cell; s^0_c is the run's own 2000–2019 state and Δs^0_c its offset from the training runs' mean (the known start state); and ϕ, λ, σ are latitude, longitude and soil code. Biomass per tree is predicted as a logarithmic ratio to its anchor, which weights relative errors equally across dense and sparse stands; the other quantities as differences. Each model has 400 trees with 31 leaves, learning rate 0.05 and 80 % row and column subsampling, and is trained on every fourth year of the training runs: five runs under nine climate models in the test folds (about three million cell-years), all six runs under all ten climate models in the deployment release (about four million).

Where the year-to-year development comes from matters for what the maps can and cannot do. In the panel version the anchor is the original's own year-by-year trajectory at that cell, averaged over its training runs and the other climate models — that is, it includes the future output of the original at that cell, but never that of the run or the climate model being predicted. The 30-year climate window moves slowly on top of it. Because no prediction is fed back, a year's error cannot carry into the next (P3); but for the same reason a single extreme year moves the window by only one thirtieth and leaves no legacy in the level. The stand is the only quantity passed from the maps to the daily model; it affects next year's fluxes but not next year's maps. In the full-grid version (Sect. 6.7) and in the nitrogen-on version (Sect. 6.6) the anchor is instead each cell's own simulated historical state, constant in time, so the climate alone carries time. Like any tree ensemble, the maps cannot extrapolate: their outputs stay within the range of the training targets, which damps responses beyond the training climates (Sects. 6.6 and 6.7).

4.4 Weather-driven departure of tree density and biomass

A forest that loses a third of its trees in a drought year does not return to its expected level the next year. To let the emulator carry such events, a second gradient-boosted model per quantity predicts, for tree density and biomass per tree, the departure

$$D^q_{c,y} = \ln \frac{X^q_{c,y} + 1}{M^q_{c,y} + 1}, \qquad D^q_{c,y} = h_q\Big(D^n_{c,y-1},\ D^a_{c,y-1},\ M^n_{c,y},\ M^a_{c,y},\ \bar{x}_{c,y},\ z_{c,y},\ z_{c,y-1}\Big)\,, \tag{6}$$

where X is the emulated value, M the forest map's, n and a stand for tree density and biomass per tree, and $z_{c,y}$ are the 25 yearly climate statistics of year y as anomalies against the 30 years before (in standard deviations). It is trained on the departures of the original's training runs from forest maps fitted without them (cross-fitted in three folds of climate models), averaged over the five runs that share each climate — the weather-forced part — and with each series' long-term mean removed, so that it changes the year-to-year course but not the long-run level. It runs forward year by year from zero in the first emulated year. Carrying two numbers per cell, it is the only recursive part of the forest maps; Sect. 6.4 shows what it does.

Table 2. Forest maps: error of each alternative setting relative to the release (median over 15 held-out cases; below 1 is better). Dense and sparse: stands with at least and fewer than five trees per patch.

setting	release	dense density	dense biomass/tree	sparse density	dense trait medians
leaves 15 / 63	31	1.010 / 0.969	1.028 / 0.982	1.067 / 0.925	1.018 / 0.986
trees 200 / 800	400	1.010 / 0.986	1.030 / 0.981	1.047 / 0.937	1.013 / 0.986
learning rate 0.025 / 0.1	0.05	1.006 / 0.982	1.026 / 0.988	1.030 / 0.923	1.013 / 0.993
no subsampling	0.8	0.985	0.995	0.985	1.006
min. 100 rows per leaf	20	0.975	1.003	0.995	1.001
climate window 10 / 20 / 50 yr	30 yr	0.988 / 0.991 / 0.999	1.017 / 1.002 / 1.036	0.874 / 0.966 / 1.008	1.005 / 1.005 / 1.004
training rows every 2nd / 8th yr	4th	0.980 / 1.000	1.003 / 0.997	0.974 / 1.009	1.007 / 1.003
without start state	with	1.005	1.011	0.973	1.294
without climate offset	with	1.012	1.112	1.136	1.026
without anchors as inputs	with	1.003	1.021	1.012	1.014

4.5 Choice of architecture and settings

Forest maps and departure model. Each setting of the gradient-boosted maps was varied on its own around the release, refitted for all five held-out climate models and scored on the 15 held-out cases as the ratio of its error to the release's (median over cases; Table 2). An alternative would have replaced a release setting had it reduced the error of both dense-stand tree density and biomass per tree by at least 2 % without worsening any quantity by more than 2 %; none did. The release sits on a plateau: fewer leaves (15), fewer trees (200) or a lower learning rate (0.025) each raise the dense-stand biomass-per-tree error by 2.6–3.0 %; more leaves (63), more trees (800) or a higher rate (0.1) each lower the dense tree-density error by 1.4–3.1 % but not the biomass error by 2 % (63 leaves: −3.1 and −1.8 %, and no quantity worse). Without subsampling, or with at least 100 rows per leaf, the errors change by less than 2.5 %. A 50-year climate window is worse (biomass per tree +3.6 %), 10 and 20 years are equivalent to 30; training rows every second or eighth year instead of every fourth are equivalent. Every input group is needed: without the climate offset from the anchor runs the biomass error rises by 11 % and the sparse-stand density error by 14 %, and the late-century drift of biomass grows (1.56 instead of 1.41); without the anchors as inputs the errors rise by 1–2 %; without the run's own 2000–2019 state the density and biomass errors are unchanged but the dense-stand trait medians are 13–41 % worse. For the departure model, 15 or 63 leaves, 200 or 800 trees and a 15- or 20-year anomaly window all changed the errors by less than 3 %; the share of a die-off it reproduces moved between 0.15 and 0.21 (release 0.18).

[**Daily networks: to be completed when the five-network tests of the two qualifying settings (four layers; batch 4 096) have run.**]

Evidence already measured:

- *Daily network size.* A two-layer network of 64 units left a worst held-out-cell NPP error of 7.9 % and area totals off by more than 5 %; three layers of 256 units reduced it to 5.7 %, at 3.3 times the cost ($1.3{\times}10^{-3}$ instead of $4.0{\times}10^{-4}$ core-seconds per cell-year and network).
- *Ensemble size.* A single network's NPP level shifts by several per cent with its random seed, and one network's error is 1.01–1.28 (worst 1.72) times that of five; three networks lost 10–11 % on evapotranspiration and runoff in one of two replicates; ten networks improved evapotranspiration by 6–9 % and runoff by 6–13 % at twice the cost, which puts the emulator at the limit of the T31 allowance. The members' spread is not used as an uncertainty estimate.
- *Training sample.* 80 million cell-days instead of 40 million changed NPP by less than run-to-run noise; 160 million biased NPP area totals by −9 to −10 %.
- *Deep store.* Without drainage the deep store accumulated 2–3 m of water over 81 years and the runoff error was 11–14 times a further run's; with $\tau = 365$ days it is 3.9–5.5 times; 30 days gave the same 30-year means and worse single years.
- *NPP as GPP minus respiration* rather than a separate output: area totals of NPP off by 2.3 instead of 5.4 %. A structured photosynthesis head with separable limiting factors fitted daily GPP 2.3 times worse.
- *Soil texture* as four further inputs left evapotranspiration unchanged (−1 %), improved runoff

by 5–11 % and NPP by 13–16 % in two replicate pairs; the released version does not yet use it.

- *The anchor.* Without an anchor the forest map agreed with the unseen run on fewer cells than a plain lookup of the training runs; with it, it beats the lookup in every case.
- *Recursion.* Feeding the map its own last-year state remembered a start error for 10–20 years but let errors build up (error growth 1.55 for tree density and 1.86 for biomass per tree, against 1.22 and 1.41).
- *Targets.* A logarithmic target for biomass per tree lowered its typical error from 1.42 to 1.26 times a further run; Poisson and Tweedie objectives for tree density only moved error between sparse and dense cells.
- *Inputs.* Leaving out latitude, longitude and soil code changed the scores by less than 3 %; the soil code as a category or as texture values by less than 1 %. Plant-type cover fitted additively beats a relative or sum-constrained fit (dense tree cover 1.55 against 1.85 and 1.58 times a further run).

4.6 Implementation

LandEmu is a Python package whose only run-time dependencies are numpy and LightGBM; the networks are ported from PyTorch to numpy (maximum difference 10^{-6} of each output's range on 20 000 real input rows). The yearly climate statistics and their anomalies equal those of the training pipeline exactly (to 4×10^{-11} for the anomalies), and a validation release reproduces the scored daily fluxes of the evaluation to rounding (median per-cell difference of 30-year means $1.5–2.9\times10^{-4}$) and the yearly quantities with a median difference of at most 2×10^{-4} of their typical size. The largest remaining yearly differences have one cause: the training table converted the forcing, stored in 0.1 °C steps, in single precision, so days at exactly 30.0 °C counted as hot days; continuous forcing from a host model never sits exactly on such a threshold. A release (models plus per-cell set-ups for 1 050 cells) occupies about 190 MB. The interface offers a whole year (`step_year`) or a day at a time (`begin_year`, 365 × `step_day`, `end_year`, bit-identical to the yearly call), so that a host model can exchange fluxes daily.

What a deployment needs follows from the anchor: every emulated cell needs a historical simulation of LPJmL-FIT, which supplies the anchors, the 2000–2019 state, the start stores and stand and the water capacity. In the panel version the anchor also uses the training runs' future output at that cell (other runs and other climate models, never the run being predicted); the full-grid and nitrogen-on versions use only the historical state.

4.7 Variant with a CO_2 input

For LPJmL-FIT with nitrogen limitation and rising CO_2, the same architecture receives the year's CO_2 concentration as an additional daily-network input, and the forest maps receive the trailing 30-year mean CO_2 and its change since 1990–2019. It is trained on runs of that model (Sect. 6.6). This variant and the full-grid version use the forest maps without the departure model.

5 Training data and evaluation

5.1 Simulations of the original

The main data set is a panel of 1 050 cells: 105 blocks of ten contiguous 0.5° cells, chosen to balance 20 climate strata (Fig. 3a). Each block was spun up for 1000 years and run over 2000–2019 under observed weather (GSWP3-W5E5; Lange et al. 2021) and over 2020–2100 under bias-adjusted output (Lange, 2019) of ten CMIP6 climate models (GFDL-ESM4, IPSL-CM6A-LR, MPI-ESM1-2-HR, MRI-ESM2-0, UKESM1-0-LL, CanESM5, CNRM-CM6-1, CNRM-ESM2-1, EC-Earth3, MIROC6) for three scenarios (SSP1-2.6, SSP3-7.0, SSP5-8.5; O'Neill et al. 2016), plus a control with observed 1990–2019 years resampled. We call each combination of climate model and scenario a *leg*. Six independent runs exist, each with its own spin-up and random seed, all with 25 patches, nitrogen limitation off and CO_2 constant after 2019. Per-tree output is written every year (trees taller than 5 m, as in the original's standard output); daily fluxes are written for 13 legs of five climate models and for the control, in runs 1–4.

5.2 Held-out tests

The forest maps are tested in folds that hold out one of five climate models entirely, together with run 4: they are trained on the other nine climate models and runs 1, 2, 3, 5 and 6, and run 4 under the held-out model is the truth (Fig. 3b). With three scenarios this gives 15 test cases. The cells themselves are seen in training, as they would be in deployment (Sect. 7.3). The daily networks are trained on runs 1–4 under all 14 legs with daily output and tested on 24 blocks (240 cells, 238 with GPP of at least $50\,\mathrm{gC\,m^{-2}\,yr^{-1}}$) never used in their training, including the five biome blocks of Fig. 2. For them, therefore, only the place is new: they have seen every climate model they are scored on. In the full emulator the daily networks read the stand predicted by forest maps that never saw the test run or the test climate model.

5.3 Measuring error against the original's own variability

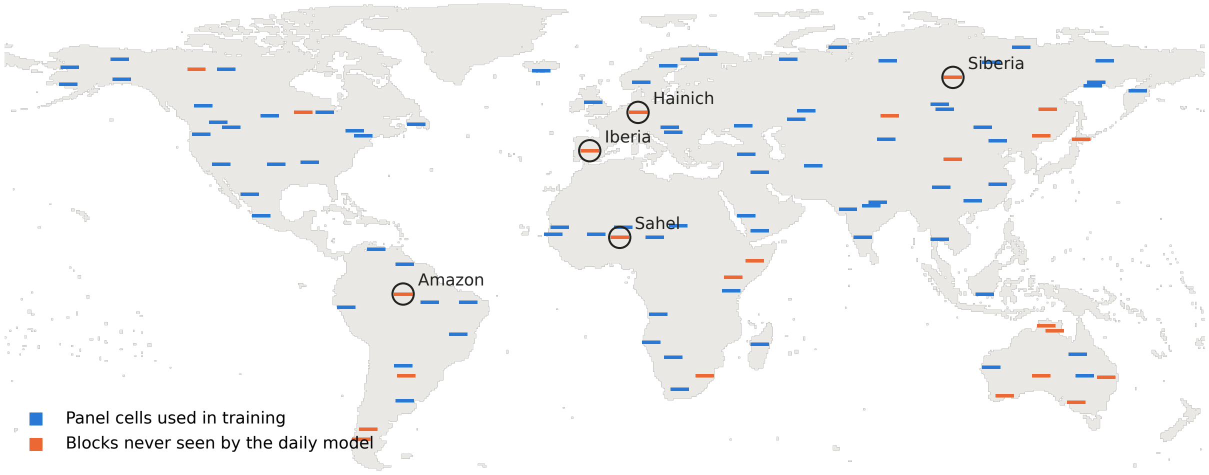
Because a second run of LPJmL-FIT differs from the first, a fixed tolerance such as "within 10 %" is uninformative: two runs of the original agree on all six structure quantities within 10 % (or within the typical two-run spread of the cell's density class, where that is larger) in only 17–21 % of cells. Following practice in climate-model emulation and model verification (Watson-Parris et al., 2022; Baker et al., 2015; Lütjens et al., 2025), we instead compare the distribution of errors over cells with that of a further run. For cell c and quantity q against the unseen truth run T, the emulator's error is $e_E(c) = |E(c) - T(c)|/|T(c)|$ and a further run R's error is $e_R(c) = |R(c) - T(c)|/|T(c)|$, pooled over the available further runs. The *error ratio* at centile k is

$$\rho_k = \frac{P_k\left[e_E\right]}{P_k\left[e_R\right]}, \tag{7}$$

with $k = 50$ (the typical cell) and $k = 90$ (the worst tenth of cells). A further run of the original scores $\rho \approx 1$ by construction; the mean of three runs scores about 0.82, and a perfect predictor of the expectation about 0.71, because an average beats a single random realisation. Values below one are therefore possible and, for a model that predicts the expectation, expected. We apply the ratio to 30-year means (2071–2100 for structure; 2021–2050 and 2071–2100 for fluxes), to the climate response (late minus early period), and to single years (for drift). Area totals (cosine-latitude weighted) are reported as signed relative errors, because a coherent bias of a few per cent can hide behind small per-cell errors yet does not average out in a sum.

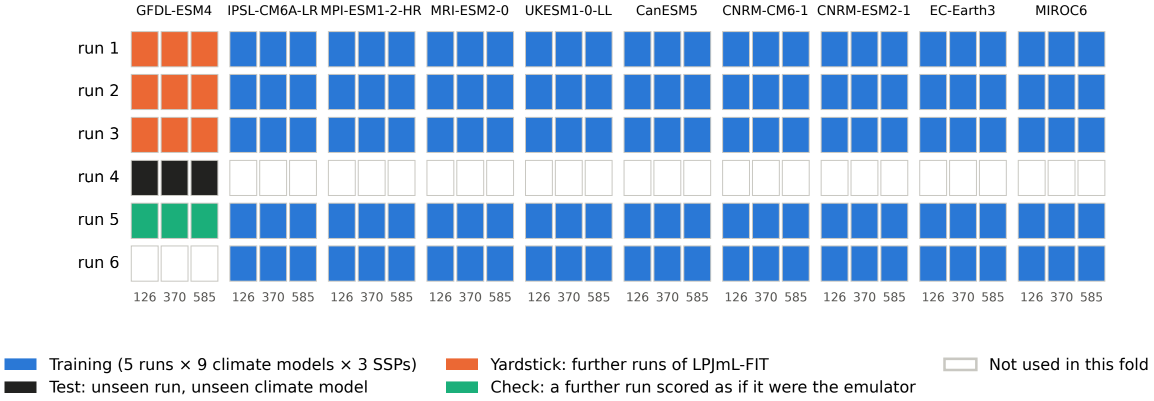
Forests with at least five trees per patch in the anchor are reported as *dense* (about 80 % of tree-bearing cells), the others as *sparse*; classing them by the anchor rather than by the truth run avoids selecting cells on the outcome being predicted. Each number is the median over the 15 test cases, with the range over cases where it matters. *Drift* means the gap to the original growing, not the values changing; we measure it as the ratio of the single-year typical error ratio late in the century (2081–2100) to that of an earlier period. For the forest structure we report two baselines: 2021–2040, when the known start state still makes the emulator's error smaller than a further run's, and 2051–2070, after that knowledge has faded.

Reference arms. Every score is reported beside (i) a further run scored as if it were the emulator, which must give about 1 and checks the scorer; (ii) the mean of other runs, a reference for a near-ideal predictor of the expectation; (iii) the anchor alone, without the learned departure; and (iv) the run's 2019 state carried forward, which shows whether the measure can detect drift.



(a) 1 050 panel cells in 105 blocks of ten (25 patches each)

Siberia
Hainich
Iberia
Sahel
Amazon
🟦 Panel cells used in training
🟧 Blocks never seen by the daily model



(b) One test fold of the yearly forest maps (climate model and run held out)

GFDL-ESM4
IPSL-CM6A-LR
MPI-ESM1-2-HR
MRI-ESM2-0
UKESM1-0-LL
CanESM5
CNRM-CM6-1
CNRM-ESM2-1
EC-Earth3
MIROC6
run 1
run 2
run 3
run 4
run 5
run 6
126 370 585
Training (5 runs × 9 climate models × 3 SSPs)
Test: unseen run, unseen climate model
Yardstick: further runs of LPJmL-FIT
Check: a further run scored as if it were the emulator
Not used in this fold

Figure 3. Training and test design. (a) The 1 050 panel cells (105 blocks of ten cells), the 24 blocks never used to train the daily networks, and the five biome blocks. (b) One of the five test folds of the yearly forest maps: one climate model (here GFDL-ESM4) and one independent run are held out; further runs under the held-out climate model give the yardstick, and one of them is scored as if it were the emulator to check the scorer.

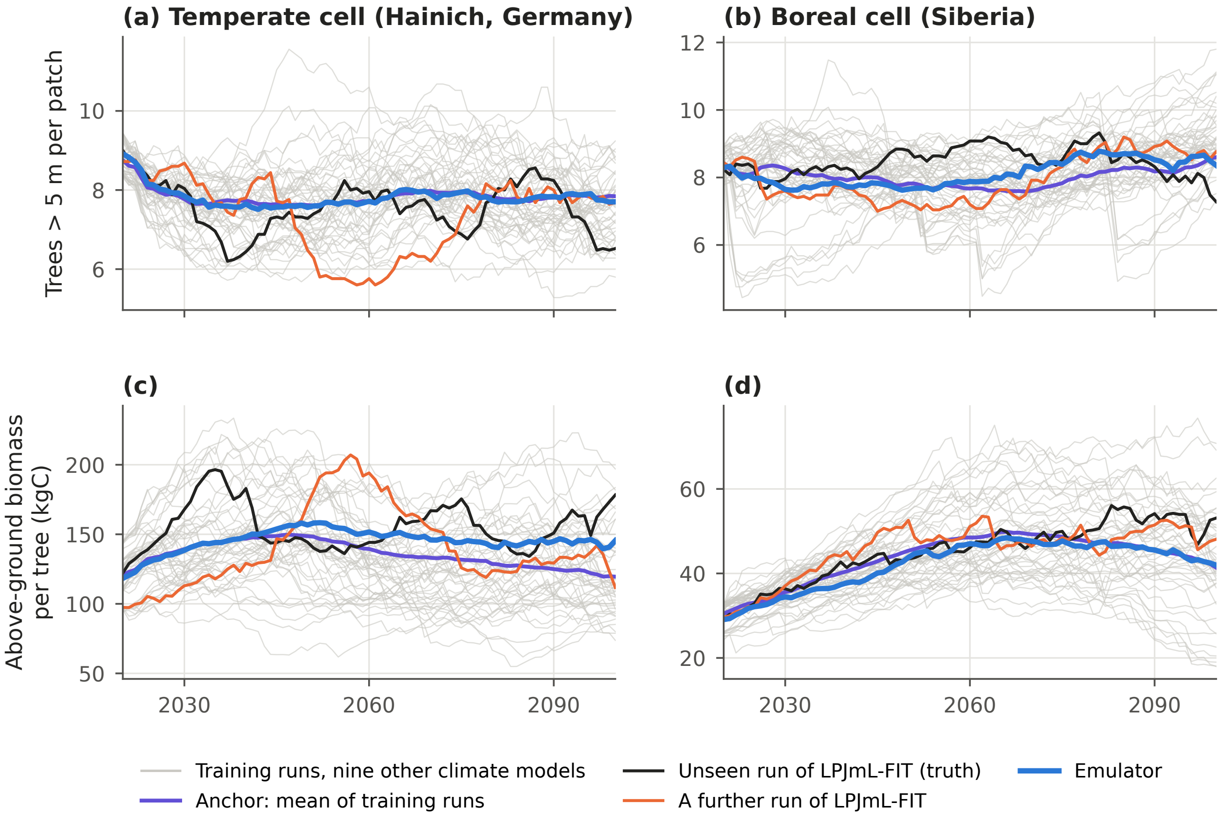


Figure 4. The yearly forest maps at two single cells under a held-out climate model (GFDL-ESM4, SSP3-7.0): (a, b) trees taller than 5 m per patch and (c, d) above-ground biomass per tree. Grey: training runs under the nine other climate models; violet: their mean, the anchor; black: the unseen run of LPJmL-FIT; orange: a further run under the same climate; blue: the emulator.

6 Results

6.1 How the forest maps behave at a single cell

Figure 4 shows what the yearly maps do. At the temperate and the boreal cell the original's runs scatter widely: its unseen run and a further run of the same climate wander apart by up to two trees per patch and by 50 % in biomass per tree for decades, because a few deaths or recruitments in 25 patches change the stand. The anchor — the mean of the training runs under other climates — smooths this noise but knows nothing about the held-out climate. The emulator follows the anchor where the climate resembles the training climates and departs from it where it does not: at the temperate cell it keeps biomass per tree above the declining anchor in the second half of the century, as the unseen run does. It does not reproduce the excursions of the single run, and it should not: they are not predictable from climate.

6.2 Forest structure

Over all 15 held-out cases (Fig. 5), the typical error of tree density in dense forests is 1.22 times that of a further run (cases 1.12–1.45), and that of biomass per tree 1.18 times (1.03–1.61). The four trait medians are closer to the unseen run than a further run is (0.59–0.86 times): the emulator predicts the expectation, while a further run is itself a random draw. In the worst tenth of dense cells tree density reaches 1.40 times and biomass per tree 1.13 times a further run. Area totals over dense cells are off by a median of −2.6 % for stems (cases −4.7 to −0.2 %) and −1.3 % for biomass (−8.6 to +2.7 %). Sparse stands are harder: tree density 2.90 times a further run, biomass per tree 1.55 times, stem and biomass totals a median −3.3 % (−14 to +12 %) and −1.5 %

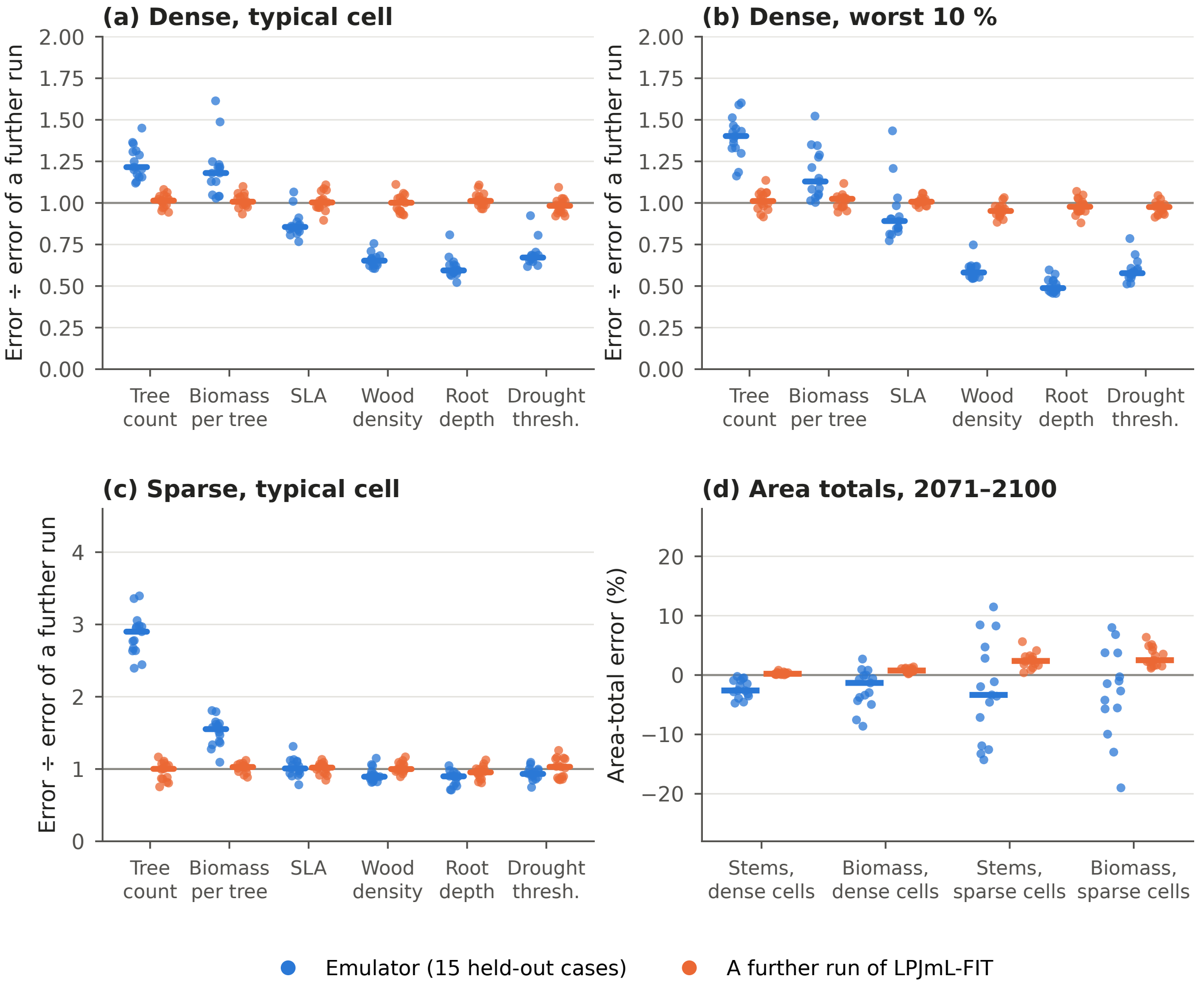


Figure 5. Forest structure in 2071–2100 under a held-out climate model and run (15 cases: five climate models × three scenarios). (a–c) Error ratio for six quantities: typical dense cell, worst tenth of dense cells and typical sparse cell; dots are cases, bars medians. Orange: a further run of LPJmL-FIT scored the same way, which must give about 1. (d) Signed area-total errors of stems and biomass.

(−19 to +8 %). The departure model of Sect. 4.4 contributes to these numbers: the forest maps alone give 1.26 and 1.18 for dense tree density and biomass per tree, 1.47 for tree density in the worst tenth, 3.00 and 1.47 in sparse stands, and dense area totals of −0.2 and −1.5 %. So the departure model lowers the tree-density errors by 3–5 % and makes sparse biomass per tree 5 % worse and the dense stem total 2.4 points more negative. Classing dense cells by the held-out run instead of the anchor gives similar ratios (forest maps alone: 1.20 for tree density and 1.16 for biomass per tree in dense cells, 3.15 for sparse tree density).

6.3 Drift of the forest state

Run year by year from 2020 to 2100 (Fig. 6), the emulator's error starts below that of a further run, because it knows the run's own 2000–2019 state, and rises over about three decades as that knowledge fades. It then levels off: tree density at 1.06–1.24 times a further run's error in the 2050s and 1.07–1.41 in the 2090s (range over cases), biomass per tree at 0.96–1.24 and 0.99–1.68. From 2051–2070 to 2081–2100 the typical error grows by a median factor of 1.05 for both tree density (cases 0.99–1.14) and biomass per tree (0.97–1.36); the trait medians grow by 1.11–1.20 while remaining at 0.63–0.88 times a further run. Measured from 2021–2040, when the start state still keeps the error low, the growth is 1.20 for tree density and 1.41 for biomass per tree. The

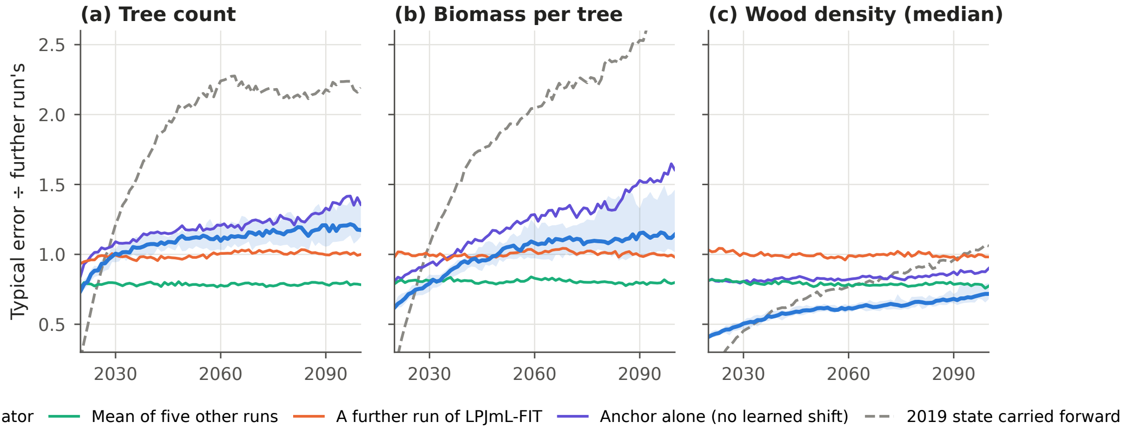


Figure 6. The forest maps run year by year: typical-cell error ratio of dense cells per year, median over the 15 held-out cases (shading: 10th–90th percentile of cases for the emulator), for (a) tree density, (b) biomass per tree and (c) median wood density. Reference arms: a further run (orange), the mean of five other runs (green), the anchor without the learned departure (violet) and the 2019 state carried forward (grey dashed).

rise in the first decades does not depend on the scenario (tree density 1.22 under SSP1-2.6, 1.23 under SSP5-8.5), which identifies it as the fading start state rather than climate extrapolation. What remains is a slow creep of tree density in the second half of the century that increases with warming (+0.05, +0.10 and +0.12 from the 2050s to the 2090s for SSP1-2.6, SSP3-7.0 and SSP5-8.5). The reference arms behave as they should: a further run stays at 1.0, the mean of other runs at 0.8, the anchor alone rises steadily to 1.3–1.6, and the carried-forward 2019 state rises to more than twice a further run's error, so the measure does detect drift.

6.4 Drought die-offs

Year by year, the original's forests do not change smoothly. Over the 15 test cases, 3 483 dense cells of the unseen run lose more than 15 % of their trees in a single year of 2025–2090 (counting each cell's largest loss), and in 94 % of these events at least two of the three further runs lose at least half as much in the same year; on average they reproduce 94 % of the loss. The losses are forced by the weather, not by chance. They are droughts: in the event year, annual precipitation is a median 1.25 standard deviations below the cell's normal, the climatic water balance 1.20 and relative humidity 1.18; the year before is dry too. Over the 3 291 such events of the 15 test cases, the original loses a median 33 % of its trees in the event year and is still 15 % below its earlier level ten years later (Fig. 7).

The forest maps alone, whose climate input is a trailing 30-year mean, follow 3 % of such a drop. With the departure model the emulator follows 18 %, and its error in the ten years after an event falls from 4.5 to 3.8 times that of a further run (stand biomass: 7 to 16 % of the drop). The same model without the weather inputs follows no more than the maps alone (3 %), so the gain comes from the weather anomaly.

What limits it is the mechanism of the die-offs. Reconstructing each tree's leaf carbon from the original's per-tree output shows that 94–98 % of the trees removed in these years had less leaf carbon than a sapling of their own specific leaf area — the threshold below which LPJmL-FIT removes a tree — and sat at a median 0.58 of it, while no tree that died by the random mortality draw did. The deaths in event years equal those expected from the trees' own mortality probabilities (34.2 against 34.3 %), so fire, which is active in these runs, does not cause them. A die-off is therefore many trees crossing an individual threshold in the same year, driven by a one-year collapse of their carbon balance. Neither the weather nor the stand describes this well: the original's own yearly NPP anomaly separates event years from other years with an area under

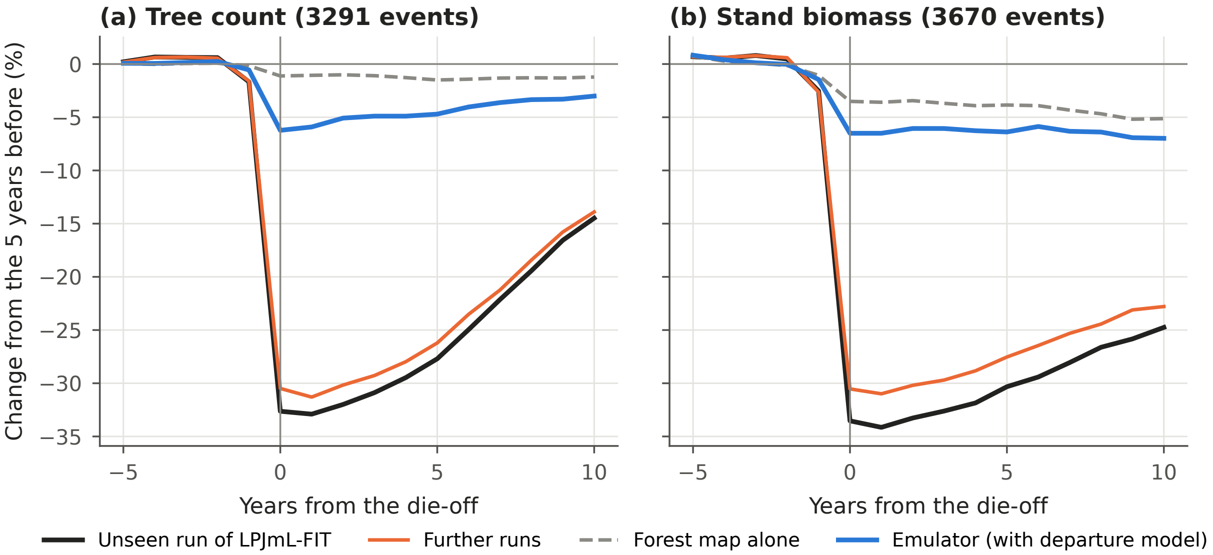


Figure 7. Drought die-offs: composite over the original's largest one-year losses that its further runs share (dense cells, 15 held-out cases), relative to the five years before. (a) Trees per patch, (b) stand biomass. Black: the unseen run of LPJmL-FIT; orange: further runs; grey dashed: the forest maps alone; blue: the emulator with the weather-driven departure model.

the receiver-operating curve of 0.90, a forcing-only daily soil-water index 0.88, and the share of trees close to the threshold the year before 0.49–0.57. Adding the daily soil-water index and runs of consecutive dry years to the departure model raised the share of the drop it follows only to 25 %, at the price of worse sparse-stand biomass.

6.5 Daily carbon and water exchange

At the three biome cells of Fig. 8 the full emulator — daily networks never trained on these cells, fed a stand from forest maps that never saw the climate model or the run — reproduces the seasonal course and the day-to-day variability of GPP and evapotranspiration: the spring onset in the temperate and boreal forests, the short wet season at the semi-arid cell, the response to individual dry spells. Year-to-year variations of annual GPP follow the original's at the temperate and semi-arid cells (correlation 0.93 and 0.94; a further run: 0.91 and 0.96), less so at the boreal cell (0.65; a further run: 0.99). The 2021–2100 means differ by +0.8 % at the temperate and −0.6 % at the boreal cell, but by +12 % at the semi-arid cell, where the original's own two runs differ by 3 %. The top-metre water store is the least faithful daily variable: at the temperate cell it dries 20 mm too deep in summer, and at the boreal cell it holds up to 50 mm more water in summer than the original, while the fluxes stay close. The networks are trained on fluxes and store changes, and the store is a state integrated by the bucket, not a prediction target.

Over the 238 held-out cells and 13 legs (Fig. 9), the typical error of 30-year mean GPP is 2.4 times a further run's (median over legs), evapotranspiration 3.1 times, NPP 3.9 times and runoff 5.2 times. Since two runs of the original differ by only 1.0–1.6 % in a cell's 30-year GPP, evapotranspiration or NPP, this means typical per-cell errors of about 3–6 % for GPP and evapotranspiration and 5–6 % for NPP. Area totals are off by a median of 1 % for GPP, NPP and evapotranspiration and 2 % for runoff, and by at most 2.9 % on any leg. The climate response (2071–2100 minus 2021–2050) is 1.8–2.4 times a further run's error. Runoff is hardest because it is a small residual of precipitation minus evapotranspiration, so the evapotranspiration error passes into it amplified; NPP is hard because it is a small difference of two large terms in the most productive cells.

The emulated stand costs accuracy. Fed the original's own stand instead, the same daily networks reach 2.2 (GPP), 2.6 (evapotranspiration), 3.4 (NPP) and 4.5 (runoff) times a further run's error. Compared leg by leg and period by period, the emulated stand adds a median 12 % to the GPP error and 37 % to the evapotranspiration error, most of it through the cover of the plant types. On these legs the gap to the original does not grow for the median of any flux: the late-to-early ratio of single-year errors is 0.90–1.04; it grows only for GPP on the two warmest legs (UKESM1-0-LL SSP3-7.0, IPSL-CM6A-LR SSP5-8.5; 1.26 and 1.28). The deep water store stays within the range of the original's change, and over 81 years the across-cell mean of the top-metre store stays between 80 and 91 mm.

In this evaluation the climate models are seen by the daily networks; only the cells are new. That matters: in tests with a climate model withheld from daily-network training entirely, the gap to the original in GPP and NPP grew by a factor of 1.3–1.4 over the century. The cause was the daily carbon response degrading as the climate moved away from the training climates, not the water stores, since a run with the stores reset daily from the original drifted identically. Adding further climate models to training did not reduce this growth beyond run-to-run noise for UKESM1-0-LL; it falls (to 1.17 for NPP) only when that model itself is in training.

6.6 Learning the warming response

The warming response is the property an ESM needs most and the one most emulators never test. We tested it on LPJmL-FIT with nitrogen limitation and rising CO_2, run on 674 cells that each represent one region of a 674-region climate classification of the land surface (25 patches each), under the ten climate models and three scenarios, with CO_2-fixed twins of every scenario for the training climate models. Two climate models of low and high climate sensitivity, MPI-ESM1-2-HR and IPSL-CM6A-LR, were held out entirely, together with 20 % of the cells; both lie within the warming range of the eight training models, of which CanESM5 and UKESM1-0-LL warm more, so the test climate is covered by the training climates. An independent second run of the original exists for the held-out legs. Here the forest maps are anchored on each cell's own 1990–2019 state, so no future output of the original enters the prediction.

With rising CO_2, the original's vegetation carbon increases by 32–45 % by 2071–2100 on the held-out legs (Fig. 10a), mostly through CO_2 fertilisation. The emulator reproduces 81–97 % of this change, while the original's independent second run reproduces 61–89 %: the original's own global response varies by up to 40 % between runs on this grid, because single cells stand for whole regions. Trained only on SSP1-2.6, the same maps reproduce 37–43 % — the failure seen when an emulator meets a climate warmer than its training climates.

With CO_2 fixed at its 2019 level, the original loses vegetation carbon under warming, by up to 20.7 % for IPSL-CM6A-LR under SSP5-8.5 (second run: 26.3 %). The emulator reproduces 72 % of this loss and of the IPSL-CM6A-LR loss under SSP3-7.0. The CO_2-fixed twins of SSP1-2.6 and SSP5-8.5 were added to training after the first held-out results, in which twins of SSP3-7.0 alone gave 67 and 57 % of these losses; the held-out models and cells were unchanged and nothing else was tuned. Contrasting warming with and without CO_2 rise at strong warming is therefore what teaches the warming-only response. Without such twins the maps predict carbon gains on all four warming-only legs, because all their training legs carry rising CO_2. Per cell (Fig. 10b) the emulator's change is as close to the original's as the second run's is (root-mean-square error 0.65–0.80 times the second run's), and the slopes of both against the original are below one (0.73 and 0.75): the original's per-cell change contains run-to-run noise, which flattens the regression slope of any predictor, including a perfect estimate of the expected response.

For GPP the picture splits. The daily networks fed the original's stand reproduce the global GPP change within 4 % and the per-cell change with slopes of 0.92–0.99, with rising and with fixed CO_2. Fed the emulated stand, the global change with rising CO_2 is still within 6 %, but the per-cell slope falls to 0.66–0.79, because the forest maps' response of leaf area and tree cover is too uniform across cells (slopes 0.73–0.83, against 0.89–0.96 for the second run). The global totals are right; the cell-to-cell spread of the response is too small. On the held-out cells, a learning curve over 135, 269 and 539 training cells reduced this error by 6–12 % per doubling, which points to

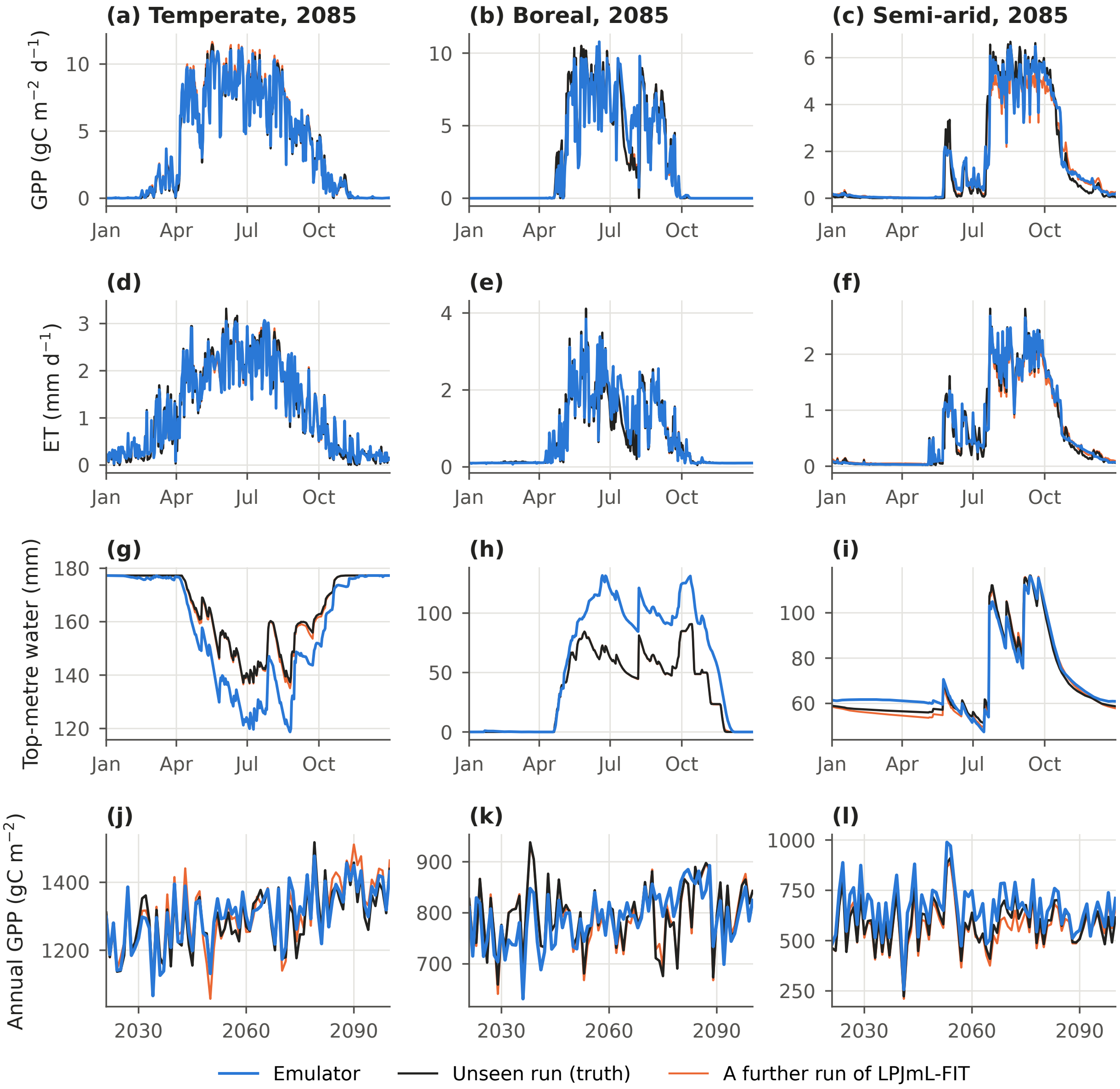


Figure 8. Daily output of the full emulator at three biome cells never used to train the daily networks (GFDL-ESM4, SSP3-7.0; forest maps with this climate model and run held out). (a–c) Daily GPP, (d–f) evapotranspiration and (g–i) top-metre soil water in 2085; (j–l) annual GPP 2021–2100. Black: the unseen run of LPJmL-FIT; orange: a further run; blue: the emulator.

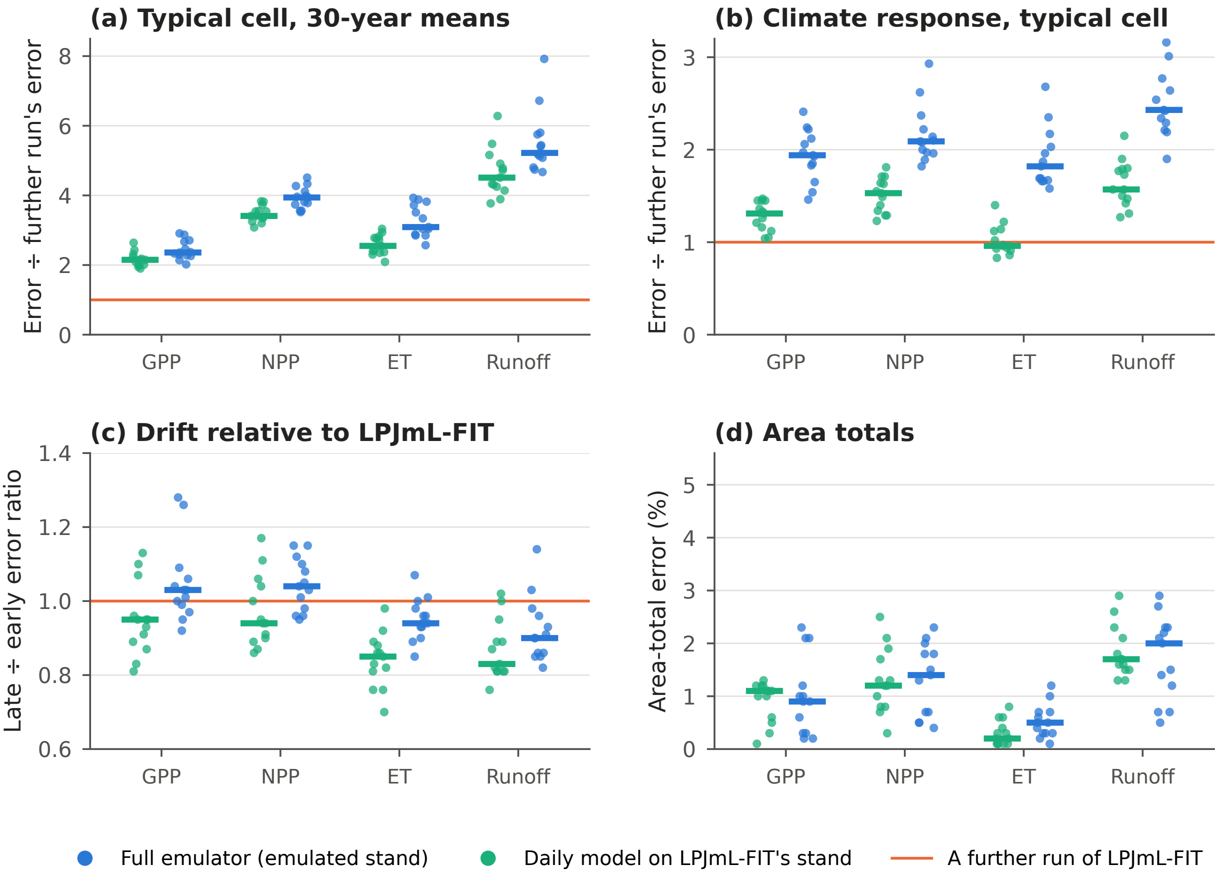


Figure 9. Daily fluxes over 238 held-out cells and 13 climate-model legs (dots: legs; bars: medians). (a) Typical-cell error ratio of 30-year means (larger of 2021–2050 and 2071–2100); (b) error ratio of the climate response; (c) drift, the late-to-early ratio of single-year errors (1 = no change in the gap to the original); (d) area-total errors. Blue: the full emulator; green: the same daily networks fed the original's stand. Orange line: a further run of the original.

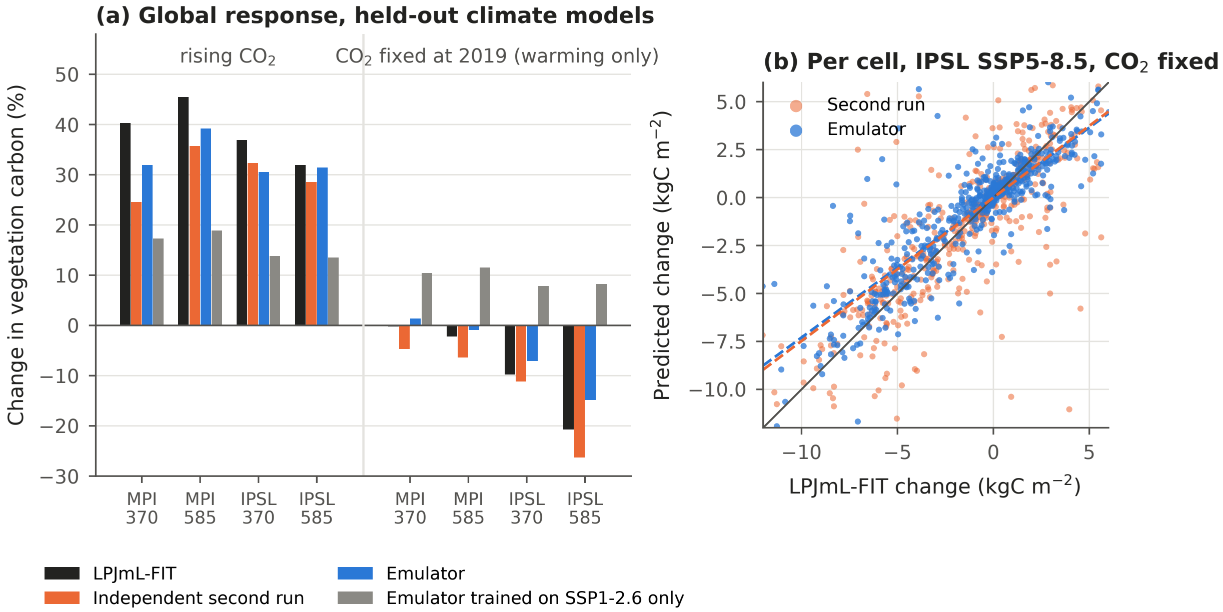


Figure 10. The warming response under two climate models never seen in training (LPJmL-FIT with nitrogen limitation, 674 representative cells). (a) Area-weighted change of vegetation carbon from 1990–2019 to 2071–2100 with rising CO_2 and with CO_2 fixed at 2019: the original, its independent second run, the emulator, and the same emulator trained on SSP1-2.6 only. (b) Per-cell change for IPSL-CM6A-LR, SSP5-8.5, fixed CO_2: emulator (blue) and second run (orange) against the original; dashed lines are least-squares slopes, the solid line is the 1:1 line.

the number of training cells rather than the model form as the limit.

6.7 Full-grid application

To show that the emulator runs globally, we retrained the forest maps on four runs of LPJmL-FIT (nitrogen off) under GFDL-ESM4 — historical, SSP1-2.6, SSP2-4.5, SSP3-7.0 — made with a different build of the model, anchored each cell on its own simulated 1985–2014 state, and emulated each future scenario with maps that never saw it. The daily networks were not retrained (these runs store no daily output). LandEmu then ran on all 64 240 land cells from 1975 to 2100 (Fig. 11). Over 1985–2014 the emulated global GPP equals the original's (+0.04 %), NPP is 1.1 % high, transpiration 8.5 % low and runoff 5.8 % high. Benchmarked against observations with the ILAMB system, the emulator's overall scores are within 0.03 of the original's on all 27 data-set comparisons; for biomass, leaf area and cover this is not a test of the emulator, because its historical forest state is anchored on the original's. Over the same period the original's GPP rises from 151 to 162 $\mathrm{PgC\,yr^{-1}}$ because CO_2 rises in its historical forcing, while the emulator's stays flat: the nitrogen-off emulator has no CO_2 input by design (Sect. 2), so it reproduces the level but not this trend.

Under the unseen scenarios, the typical per-cell error in 2071–2100 is 6.0–6.4 % for GPP, 4.4–4.8 % for evapotranspiration, 11–15 % for vegetation carbon and 7–8 % for dense-cell tree density, always smaller than when the 1985–2014 state is carried forward (11–15 %, 7–12 %, 18–26 % and 10–11 %). The global responses agree in sign in 13 of 15 cases (five quantities, three scenarios), but the carbon responses are 3–5 percentage points too low, and under SSP3-7.0, the warmest scenario and the only one outside the training range, the emulator captures only a third of the original's vegetation-carbon loss (−2.8 against −9.2 %). This is the extrapolation limit of Sect. 6.6 reproduced on the full grid: with only milder scenarios in training, the strongest response is damped. On an earlier set of eight independent runs of this model under GFDL-ESM4, scored over all 57 726 tree-bearing cells against a further run, an earlier form of the forest maps that predicts 30-year windows rather than single years reaches 0.92 times a further run's error for dense

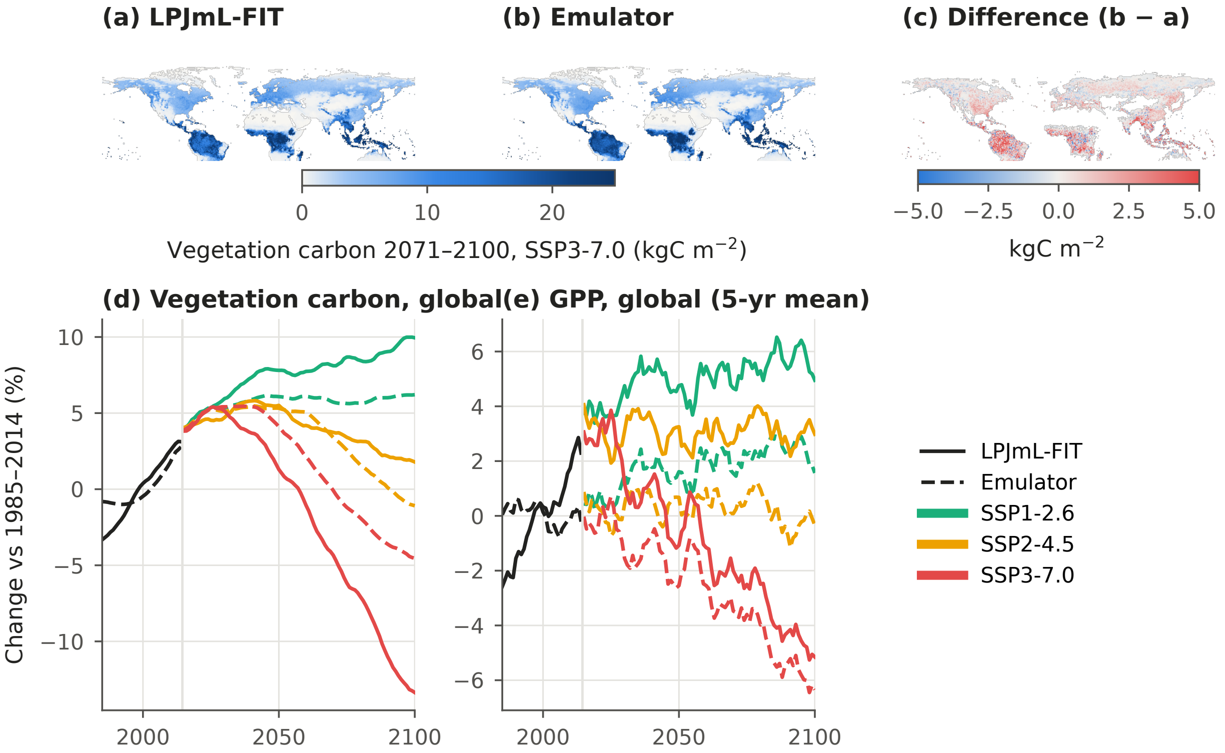


Figure 11. LandEmu on the full 0.5° grid. Vegetation carbon in 2071–2100 under SSP3-7.0 in (a) LPJmL-FIT and (b) the emulator, whose forest maps never saw SSP3-7.0, and (c) their difference (cells with at least 0.5 kgC m^{-2}). (d) Global vegetation carbon and (e) global GPP as change relative to the original's 1985–2014 mean (5-year running means): LPJmL-FIT solid, emulator dashed. The historical forest state is anchored and therefore not a test.

tree density and biomass per tree when only the run is held out, and 1.24 and 1.47 times when the scenario is held out as well.

6.8 Cost

Measured end to end on one core over 1 050 cells and 81 years, the whole emulator costs 7.0×10^{-3} core-seconds per cell-year (Fig. 12), 97 % of it in the daily networks. LPJmL-FIT, timed on one core of the same computing cluster at the temperate cell (marginal rate), costs 0.267 core-seconds at 25 patches and 7.06 at 500 patches, so the emulator needs 38 times fewer core-seconds than the original at its training setting and about 1 000 times fewer than at the publication setting. The original's cost depends on how many trees a cell carries: across the five biome cells it is 0.20–0.33 core-seconds at 25 patches, which gives factors of 29–47. The emulator lies inside the allowance for an intermediate-resolution atmosphere by a factor of 1.9 (T31) to 4.3 (T63). The nitrogen-on variant costs 7.3×10^{-3} core-seconds, 115–130 times less than that model on the same cells. Training is cheap as well: about 15 GPU-minutes per network and seconds to minutes per forest map. The cost is dominated by one matrix product per day and network, so batching over years or a GPU would reduce it further; neither is used here.

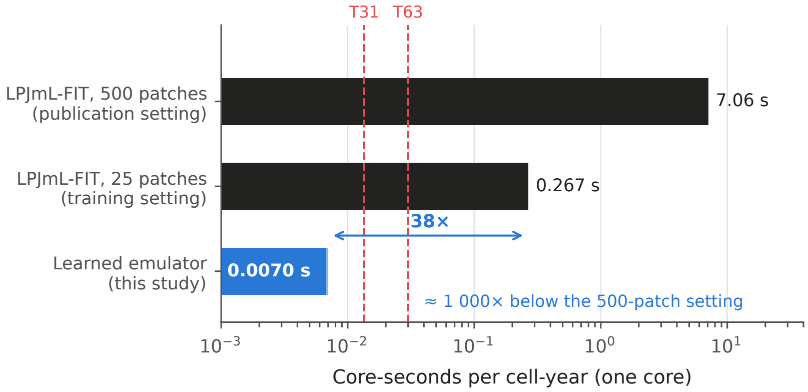


Figure 12. Cost per cell-year on one core (marginal rate): LPJmL-FIT at its training and publication settings and LandEmu (dark blue: daily networks, 97 %; light blue: yearly statistics and forest maps). Dashed: land allowances for T31 and T63 atmospheres (10 % of a coupled atmosphere's cost).

7 Discussion

7.1 What the design buys

Three design choices carry most of the result. *Learning the whole daily step* is what makes the emulator fast: the large speed-ups in the literature come from replacing an expensive iterative solver or a whole model step (Wesselkamp et al., 2025; Yang et al., 2026; Hemmen et al., 2026), and the profile of LPJmL-FIT shows that nothing short of this can give an order of magnitude for an individual-tree model. *Predicting the expectation* removes the patch ensemble, which is a variance-reduction device of the original, not a process; at the publication setting it is worth more than every process combined. *Keeping the forest level non-recursive* sidesteps the error accumulation that made a one-step model of tree counts unusable and that rollout losses (Raoult et al., 2026; Wang et al., 2023) or trajectory fitting (Ekholm et al., 2024) address in other emulators; the only recursive part of the forest maps is a bounded, weather-driven departure from that level. The daily stores are recursive too, but they are closed physical stores rather than a learned latent state, and they did not walk away from the original in 81-year free runs. Table 3 sets LandEmu against the properties an ESM land component needs (Sect. 1).

7.2 What limits the fidelity

Four findings locate the remaining error. First, the emulator is *data-limited in the space of climates, not in the number of runs*. On the panel, adding five climate models to training raised the share of cells close to the unseen run by 0.023 and lowered the biomass-per-tree error ratio from 1.64 to 1.42, while adding two runs of the original raised it by only 0.005. The warming-response test makes the same point from the other side: with training climates that cover the test climate the response is reproduced, with only milder climates it is damped by more than half. Second, *sparse stands* carry most of the structure error. Reweighting the loss towards them only moved error between sparse and dense cells, while the mean of three runs is 0.82 times a further run in every density class — so their response is predictable in principle, and what is missing is information about how sparse cells respond. Third, *the year-to-year course is a property of individual trees*: the original's drought die-offs are threshold kills of single trees, decided by each tree's leaf carbon in a year of carbon collapse, which no stand-level quantity singles out sharply. Fourth, the daily networks' own error, two to three times a further run, still exceeds the 10–40 % that the emulated stand adds, but the stand's share is concentrated in evapotranspiration and in the cell-to-cell spread of the climate response.

Table 3. LandEmu and the properties an ESM land component needs.

Property	**LandEmu**
Prognostic state and memory	explicit snow, top-metre and deep water stores; trailing weather windows; 30-year climate window; stand carried yearly; a two-number weather-driven departure of the forest state. No learned latent state.
Coupling interface	daily latent-heat-relevant fluxes (interception, transpiration, soil evaporation), runoff, snow; GPP and NPP; leaf area and plant-type cover from which albedo and roughness can be derived. Not provided: energy fluxes and surface temperature (LPJmL-FIT computes none), heterotrophic respiration and fire, hence the net CO_2 flux.
Conservation	water closed exactly every day by construction. Carbon is not tracked between NPP and vegetation carbon, which are predicted separately.
Stability when run freely	structure error grows by a factor of 1.05 after the start state has faded, with a slow creep of tree density under strong warming; flux error stable on climate models seen in training except GPP on the two warmest legs; day-by-day interface identical to the yearly call. Feedback with a host model not tested here.
Out-of-distribution climate	yearly maps: climate model and independent run held out; daily networks: places held out, climate models seen (with one withheld, the GPP and NPP gap grew 1.3–1.4-fold); warming response reproduced globally under unseen climate models covered by the training climates, damped beyond them (one third of the loss).
Uncertainty and interface	errors expressed relative to the original's run-to-run variability; deterministic prediction of the expectation, no predictive distribution; numpy + LightGBM package with daily and yearly calls.

7.3 Limitations

- **The original must have been run on every emulated cell.** The forest maps depart from a per-cell anchor, so a cell never simulated by LPJmL-FIT cannot be emulated. On the response test, cells held out of training reach 1.8–2.2 times a second run's error for leaf area and cover, against near-parity on training cells; leaving out latitude, longitude and soil code changed this by less than 5 %, so the maps do not rely on recognising places, and the error fell by 6–12 % per doubling of the training cells. Unseen places are a matter of training-set size.
- **The structure tests hold out the climate model and the run, not the place**, as in deployment on the cells the original was run on.
- **The daily networks have seen every climate model they were scored on.** With a climate model withheld, their GPP and NPP gap to the original grew 1.3–1.4-fold over the century. NPP and runoff are the least accurate fluxes (3.9 and 5.2 times a further run's error), and the climate response of every flux is 1.8–2.4 times a further run's error.
- **Soil.** The daily networks see the soil only through its top-metre water capacity, not its texture; how fast water moves through the soil, which matters most for runoff, is therefore learned only implicitly.
- **Extrapolation.** Gradient-boosted trees cannot predict outside the range of their training targets, and under a scenario warmer than all training scenarios the response is damped (Sect. 6.7); deployment should keep the training climates broader than the target climate.
- **Die-offs.** The emulator follows 18 % of the original's drought die-offs.
- **Scope of the outputs.** The structure targets cover trees taller than 5 m, as the original's standard output does, and the traits are represented by their medians, not their full distributions. There are no energy fluxes and no net CO_2 flux.
- **Scale of the evaluation.** The detailed evaluation uses 1 050 of 54 020 tree-bearing cells; the full-grid application has one run per scenario, so its errors cannot be set against a further run.

7.4 Outlook

The day-by-day interface makes LandEmu callable from an atmosphere model; experiments with Earth system models will be reported separately. The most direct improvements are training on a full-grid set of runs that covers warmer climates (which removes the unseen-cell limit and the extrapolation problem together), soil texture as a daily input, explicit litter and soil carbon pools so that the net CO_2 flux is complete, and a calibrated predictive spread scored against several independent runs of the original. Following the original's die-offs further needs a representation of how the year's carbon balance moves the leaf carbon of groups of trees against their threshold, for example a small cohort model driven by the emulated NPP.

8 Conclusions

LandEmu emulates an individual-tree, trait-based DGVM as a whole: its daily carbon and water exchange and its yearly forest structure, trait composition and stand, from daily weather alone, with water conserved exactly. Its design — learn the whole daily step, predict the expectation over the patch ensemble, predict the forest level from climate rather than from last year's state, train on climate contrast and judge against the original's own variability — follows from where the original spends its run time and from what the literature shows to fail. Under a held-out climate model and run, dense forests are reproduced close to the original's run-to-run variability (tree density 1.22, biomass per tree 1.18 times a further run's error; trait medians better than a further run), and the structure error grows only slowly once the start state has faded. The original's drought die-offs are threshold kills of individual trees, of which the emulator follows a fifth. On places held out from their training, the daily networks reproduce GPP and evapotranspiration to a few per cent per cell; under an unseen climate model their carbon fluxes drift. Trained on climates that cover the test climate, the emulator reproduces the change in vegetation carbon under rising CO_2 as well as a second run of the original does on the global total, but it damps the largest warming-only losses, the cell-to-cell spread of the response and the response of sparse stands. At 7.0×10^{-3} core-seconds per cell-year it costs 38–1 000 times less than the original and fits the land budget of an intermediate-resolution Earth system model.

Code and data availability

The emulator package (`python/src/lpjmlfit_emulator/landemu`), the training, scoring and release scripts, the scripts that produce every figure of this paper (`scripts/paper`) and the records behind every number are in the project repository `https://github.com/rimajj/LPJmLFIT_Emulator` (to be archived on Zenodo on acceptance); this paper describes release v2. The LPJmL-FIT simulations and the released emulator (models and per-cell set-ups) are stored at the Potsdam Institute for Climate Impact Research and will be archived with the code.

Author contributions

JP conceived the study, defined its goals and made the design decisions. The implementation, simulations, analysis, figures and drafting were carried out by an AI research assistant (Claude, Anthropic) under JP's direction, with every decision recorded in the repository; JP is responsible for the final content.

Competing interests

The author declares no competing interests.

Acknowledgements

Computations used the PIK high-performance computing cluster. We thank Maik Billing for the global LPJmL-FIT simulations used in Sect. 6.7, and the LPJmL-FIT developers for the model.

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